DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

DATA SCIENCE

Submitted in partial fulfillment for the award of degree(*21INT82*)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

MEGHANA M D – 1DS22AI401

Conducted at

TechCiti Software Consulting Private Limited,
Bangalore

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangelore-78

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CERTIFICATE

This is to certify that the Internship titled **"Data Science"** carried out by **Meghana M D [1DS22AI401]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

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Signature of Reporting Head
Ms. Kasturi
Techciti Software

____

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI&ML
DSCE

____

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI&ML
DSCE

External Viva:

Name of the Examiner	Signature with Date
1) ____________________	________________
2) ____________________	________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

D E C L A R A T I O N

I, MEGHANA M D**,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **TECHCITI SOFTWARE PVT.LTD**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor’s degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:12/04/2025 USN: 1DS22AI401

Place: Bangalore NAME: Meghana M D

OFFER LETTER

TechCiti Software Consulting Private Limited.
CIN: U72900KA2018PTC117376 **D-U-N-S No. :** 86 14 54180

No. 22 23 24 25/101, BNR Complex, J.P. Nagar, Bengaluru, Karnataka 560078.
Landline: 080 4162 8482 **Email**: info@techcitisoftware.in **Website:** www.techcitisoftware.in

Ref.No. TSCPL/2024-2025/HRD/INT 7811
Date: 20th January, 2025

Internship Offer Letter

Dear **Meghana M D,**

It's our great pleasure to inform you that you have successfully qualified the interview session conducted by our company. Hence, you have been offered for the position of **"Software Developer – Intern"** (Domain – **Data Science**) Start From 21-01-2025 to 20-05-2025 Your position in Bangalore, Karnataka.

In addition to the offer, you will not receive any kind of company employment benefits, as per our company policy, while you are working as an intern.

You need to submit the below mentioned documents.

a. Two passport size photo
b. Photo ID proof (1 photocopy)
c. Photo copies of mark sheets – Matriculation /Intermediate /UG/PG
d. College Proposal Letter/College ID card photocopy

Sincerely,

TECHCITI SOFTWARE CONSULTING PRIVATE LIMITED
BANGALORE

Manager
Human Resources Department
TechCiti Software Consulting Private Limited.

ACKNOWLEDGEMENT

This internship is the result of continuous guidance, support, and encouragement from several individuals, to whom I am deeply grateful. I take this opportunity to express my sincere appreciation to everyone who contributed to the successful completion of this internship.

I would like to express my heartfelt thanks to **Dr. B G Prasad**, Principal, Dayananda Sagar College of Engineering, for providing the necessary facilities and support to undertake this internship.

My sincere gratitude goes to **Dr. Vindhya P Malagi**, Head of the Department – Artificial Intelligence & Machine Learning, for providing me the opportunity to pursue this internship and for her valuable guidance and encouragement throughout.

I am thankful to **Prof. Kavya D N**, Internship Coordinator and Assistant Professor – AI & ML, for her continuous support, timely suggestions, and motivation during the internship period.

I would also like to extend my thanks to **Prof**. **Ramya K** internal guide, for her consistent guidance, timely feedback, and unwavering support throughout the internship. Her mentorship helped me stay aligned with both academic and industry expectations.

I would like to thank **Neela Santosh**, Senior HR, Codtech IT Solutions, for the valuable industry insights and mentorship provided during my role as an intern in Machine Learning.

My thanks also go to all the faculty members and non-teaching staff of the Department of AI & ML for their cooperation and assistance during the internship period.

Meghana M D [1DS22AI401]

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CHAPTER 1

INTRODUCTION

In the contemporary era, data has emerged as the backbone of decision-making processes, revolutionizing the way businesses operate and engage with their customers. The real estate sector, being a vital component of the economy, is no exception. It thrives on dynamic market trends, customer preferences, and macroeconomic factors that demand careful and data-driven analysis. Recognizing this need, TechCiti Software Pvt. Ltd. has embarked on an innovative journey to harness the power of data science for real estate price prediction, offering a cutting-edge solution that empowers stakeholders to make informed and strategic decisions.

TechCiti Software Pvt. Ltd. is a forward-thinking technology company committed to delivering high-quality software solutions that cater to various industries. With a vision to leverage advanced analytics and machine learning, the company aims to address the complexities and challenges faced by the real estate sector. The real estate price prediction project represents TechCiti's dedication to innovation and its mission to bridge the gap between data and actionable insights.

Project Overview

The real estate price prediction project is designed to predict the prices of residential and commercial properties using advanced data science techniques. By integrating various data sources such as property features, location-specific information, historical prices, and economic indicators, this project aims to create a robust predictive model that accurately estimates property prices.

- **Data Collection and Integration**: Gather data from various reliable sources, including property listings, government databases, and market reports.
- **Feature Engineering**: Identify and engineer key features that significantly impact property prices, such as location, property size, age of the property, and surrounding amenities.
- **Model Development**: Utilize machine learning algorithms such as linear regression, decision trees, random forests, and gradient boosting techniques to build and refine predictive models.
- **Model Evaluation**: Validate the models using performance metrics like mean squared error (MSE), mean absolute error (MAE), and R-squared to ensure reliability and robustness.
- **Deployment**: Develop a user-friendly interface or API that allows stakeholders to input property features and receive price predictions in real-time.

CHAPTER 2
ABOUT THE COMPANY

TechCiti Software Pvt. Ltd. is a Bengaluru-based technology company specializing in end-to-end IT infrastructure solutions, software development, and digital transformation services. Established in 2013, TechCiti has grown into a trusted partner for businesses across India, delivering customized solutions that align with clients' strategic objectives.

Core Offerings:

- **IT Infrastructure Solutions:** TechCiti provides comprehensive IT infrastructure services, including data center solutions, networking, and cybersecurity, ensuring reliable and efficient operations for clients.
- **Software Development:** The company develops custom software applications tailored to specific business needs, leveraging modern technologies to enhance productivity and user experience.
- **Digital Transformation:** TechCiti assists organizations in their digital journey by offering services such as cloud integration, enterprise resource planning (ERP) systems, and customer relationship management (CRM) solutions.

Mission and Vision:

Mission: To achieve a leading position as a distinguished end-to-end information technology infrastructure and service provider, fostering growth through superior customer service, innovation, quality, and commitment.

Vision: To enable individuals and organizations to realize their potential by redefining their engagement with technology, thereby shaping the future.

Internship and Career Opportunities:

TechCiti offers internship programs throughout the year, targeting undergraduate and postgraduate students in fields such as IT infrastructure, finance, sales, and human resources. These internships, lasting between 4 to 24 weeks, provide students with industry exposure and practical experience, preparing them for careers in the information technology sector.

Driven by a clear mission and a customer-first approach, Techciti software pvt.ltd aims to enhance online visibility, optimize digital operations, and safeguard digital assets of businesses. The company’s growth-oriented culture, innovative service delivery, and focus on real-world problem-solving make it an ideal environment for learning, experimentation, and professional development.

As an intern, working with Techciti software pvt.ltd provided not only technical exposure but also insight into the operations of a modern tech company that is shaping the future of digital transformation for small businesses.

CHAPTER 3
SCOPE OF THE WORK

The scope of the work in this real estate price prediction project encompasses the full data science lifecycle, ensuring a comprehensive, accurate, and robust solution that addresses the real-world complexities of property valuation. As a data science initiative under TechCiti Software Pvt. Ltd., the project will cover the following key areas:

1. Data Collection and Preprocessing
    - Data Sourcing: Identify and gather data from multiple sources, including property listing websites, government real estate registries, economic indicators, and location-specific data.
    - Data Cleaning: Handle missing values, inconsistent entries, and outliers to prepare a clean and reliable dataset.
    - Data Integration: Combine data from various sources into a unified dataset to enable holistic analysis.
2. Exploratory Data Analysis (EDA)
    - Conduct thorough EDA to understand data distribution, relationships among variables, and key drivers of property prices.
    - Visualize data trends and patterns using graphs, charts, and maps to uncover insights.
3. Feature Engineering
    - Feature Selection: Identify the most relevant features impacting property prices, such as square footage, number of bedrooms, property age, proximity to amenities, and neighborhood attributes.
    - Feature Creation: Create new features that better capture the underlying patterns in the data (e.g., price per square foot, distance to transportation hubs, and property condition scores).
    - Categorical Encoding: Encode categorical variables like location or property type using techniques such as one-hot encoding or label encoding.
4. Model Development and Selection
    - Experiment with various machine learning algorithms including:
        - Linear regression for baseline models.
        - Decision trees and random forests for capturing non-linear relationships.
        - Gradient boosting models (e.g., XGBoost, LightGBM) for advanced performance.

- Perform hyperparameter tuning and cross-validation to optimize model performance.

5. Model Evaluation
    - Evaluate the model using suitable performance metrics:
        - Mean Absolute Error (MAE) for interpretability of average error.
        - Root Mean Squared Error (RMSE) to penalize larger errors more heavily.
        - R-squared to measure variance explained by the model.
    - Conduct residual analysis to ensure the model's generalizability and reliability.
6. Deployment and Integration
    - Model Deployment: Develop an API or web application interface that enables end-users to input property features and obtain real-time price predictions.
    - Integration with TechCiti Platforms: Integrate the model into TechCiti's existing software ecosystem, ensuring seamless usage for internal teams or client-facing products.
7. Continuous Improvement and Monitoring
    - Establish processes for monitoring the model's performance over time.
    - Implement feedback loops to refine the model as new data becomes available, ensuring ongoing accuracy.
8. Documentation and Reporting
    - Provide clear and comprehensive documentation covering:
        - Data sources and processing pipelines.
        - Model architecture, assumptions, and limitations.
        - User guidelines for model deployment.
    - Deliver insights and summary reports to stakeholders, highlighting key trends and findings.

CHAPTER 4

LEARNING OBJECTIVES

1. Understanding Supervised Learning through Decision Tree Implementation

The primary learning objective of this task was to understand the fundamentals of supervised machine learning, particularly decision tree classifiers. I aimed to learn how decision trees make hierarchical decisions based on feature values, how entropy and information gain influence the splitting process, and how to interpret tree visualizations. Additionally, I wanted to gain hands-on experience with data preprocessing and model evaluation in Scikit-learn, and to develop an intuitive understanding of overfitting, pruning, and decision boundaries.

2. Building NLP Pipelines for Sentiment Analysis

The goal of this task was to gain practical skills in natural language processing and sentiment classification. I learned how to clean and preprocess text data using standard NLP techniques, how to convert text into numerical form using TF-IDF vectorization, and how to apply machine learning models like Logistic Regression to classify sentiments. This task also helped me understand the challenges in dealing with unstructured data, and how to evaluate classification models effectively using accuracy, precision, recall, and F1-score.

3. Exploring Deep Learning with Image Classification using CNN

The third task aimed to provide hands-on experience with deep learning and computer vision. My key learning objectives were to understand the structure and function of Convolutional Neural Networks (CNNs), learn how to build and train them using TensorFlow, and explore techniques like data augmentation and dropout. This task enhanced my knowledge of how CNNs can automatically learn spatial hierarchies from image data.

4. Personalization Techniques using Recommendation Systems

For this task, the objective was to understand the core principles behind recommendation engines used in real-world platforms. I learned how **collaborative filtering** works, how to build **user-item interaction matrices**, and how to apply similarity metrics and matrix factorization techniques to generate meaningful recommendations. This task also improved my understanding of sparsity issues in recommender systems and evaluation techniques such as RMSE and precision-based metrics to assess recommendation quality.

CHAPTER 5

CHALLENGES AND SOLUTIONS

1. Missing values and categorical data

A key challenge during this task was dealing with missing values and categorical data, which required preprocessing before they could be used in the Decision Tree model. Additionally, tuning the depth of the tree was challenging, as deeper trees tended to overfit while shallow trees did not capture enough complexity in the data. I utilized label encoding for categorical variables and mean imputation for missing values to prepare the data. I then applied cross-validation to determine the optimal depth of the tree, ensuring it was complex enough to capture patterns without overfitting. I also implemented pruning techniques to reduce overfitting.

2. Noisy text data

A major challenge here was handling the noisy text data—such as slang, misspellings, and abbreviations—which can significantly affect model performance. Another hurdle was determining the best approach for transforming text into numerical features for the sentiment analysis model. To clean the data, I used NLTK to remove stopwords, tokenize text, and apply stemming. For vectorization, I initially explored Count Vectorizer, but ultimately used TF-IDF to capture the importance of terms in relation to the entire dataset. Hyperparameter tuning using GridSearchCV helped optimize the model, improving accuracy.

3. Long training time

The most significant challenge in this task was the long training time required to train a CNN on image data. Additionally, the model tended to overfit the training data, which resulted in poor performance on unseen test data. I reduced training time by using image resizing and GPU acceleration available through platforms like Google Collab. To combat overfitting, I incorporated data augmentation, dropout layers, and early stopping in the model to enhance generalization and improve performance on the test dataset.

4. Data Sparsity

The recommendation system task faced issues related to data sparsity, as most users had limited interaction with the items, making it challenging to calculate meaningful similarities. Moreover, understanding and implementing matrix factorization techniques required an in-depth grasp of linear algebra concepts. To address the sparsity issue, I filtered out users and items with very few ratings, ensuring that the data fed into the system was dense enough for accurate similarity calculations. I implemented Singular Value Decomposition (SVD) and used the Surprise library to apply matrix factorization techniques effectively. Additionally, I deepened my understanding of matrix decomposition through documentation and hands-on experimentation.

CHAPTER 6
ACHEIVEMENTS AND CONTRIBUTION

1. Contribution to Decision Tree Implementation and Visualization

In this task, I successfully built and visualized a Decision Tree model, contributing to the development of a predictive solution for a business problem. By implementing data preprocessing techniques and optimizing the model for performance, I was able to demonstrate the effectiveness of decision trees in classifying data and predicting outcomes. My contribution also included providing valuable insights into model interpretation through visualization, which assisted the team in understanding the decision-making process of the model.

2. Enhancing Sentiment Analysis Accuracy through NLP Techniques

For the sentiment analysis task, I was able to build a robust NLP pipeline that accurately classified customer reviews. I contributed to improving the model's accuracy by experimenting with different text preprocessing techniques and vectorization methods. Additionally, my analysis and evaluation of the model's performance using metrics like precision, recall, and F1-score helped refine the model for better real-world application.

3. Advancing Image Classification with CNNs for Improved Performance

During the image classification task, I contributed by designing and implementing a Convolutional Neural Network (CNN) for image classification. My contribution included experimenting with different CNN architectures and applying techniques like data augmentation and early stopping to improve model performance. By fine-tuning the model and analysing its accuracy, I was able to contribute to the creation of a functional model capable of high-performance image classification, which can be useful for various computer vision applications.

4. Building a Scalable and Personalized Recommendation System

For the recommendation system, I contributed by developing a functional system using collaborative filtering and matrix factorization techniques. My work involved implementing user-item interaction matrices and applying matrix factorization to generate personalized recommendations. By addressing the challenges of data sparsity and ensuring the model's scalability, I was able to contribute to the company's ability to offer personalized product recommendations, enhancing the user experience for customers.

CHAPTER 7
REFLECTION AND ANALYSIS

Throughout my internship at CodTech IT Solutions, I encountered an enriching and challenging environment that pushed me to grow both professionally and personally. Each task I worked on, from decision tree implementation to sentiment analysis, image classification, and recommendation systems, not only deepened my technical expertise but also allowed me to develop problem-solving, critical thinking, and communication skills. The real-world application of machine learning algorithms, AI techniques, and data science tools provided me with valuable insights into their potential impact on business processes and decision-making.

In the process of working on these tasks, I learned how to effectively manage large datasets, clean and preprocess them, and apply the right techniques for different types of problems. Whether it was dealing with missing data, handling noisy text, or optimizing models to reduce overfitting, every challenge offered a learning opportunity. I also learned how to critically analyze models' performance, make informed decisions on model selection, and evaluate their impact using appropriate metrics. The hands-on experience in utilizing advanced libraries such as scikit-learn, TensorFlow, PyTorch, and NLTK has significantly improved my technical proficiency.

Moreover, the internship provided an opportunity to learn how to collaborate within a team, communicate technical concepts effectively, and adapt to a fast-paced work environment. I faced some setbacks, such as understanding complex algorithms like matrix factorization for the recommendation system or tuning deep learning models for image classification, but these challenges led me to become more resourceful and resilient. I learned to consult various resources, ask insightful questions, and iterate on solutions to achieve the desired results.

This experience has also heightened my understanding of how data science can solve real-world problems, from automating processes to enhancing customer experiences through personalized solutions. It has reaffirmed my passion for AI and data science and has motivated me to continue learning and improving in this field. Overall, this internship has been a transformative journey, equipping me with technical, problem-solving, and professional skills that will undoubtedly support my future career in data science and AI.

CHAPTER 8
RECOMMENDATIONS

Based on my experience, I recommend incorporating structured mentorship for interns. Regular feedback sessions with experienced professionals would help interns track their progress and refine their skills. A mentor-focused approach can also provide clarity on specific tasks and career development.

Introducing workshops or specialized training sessions on advanced machine learning, AI, and emerging technologies would be beneficial. These workshops could provide interns with in-depth knowledge and practical skills in areas such as deep learning, model optimization, and data science trends.

I suggest fostering cross-functional collaboration between interns and different teams. Allowing interns to work alongside professionals from various departments, such as data science and development, would provide a holistic view of project execution and promote valuable teamwork skills.

Incorporating project-based challenges or hackathons would encourage innovation and practical problem-solving. These challenges simulate real-world scenarios and provide interns the opportunity to showcase their abilities, pushing them to apply their learning in creative ways.

Lastly, enhancing the documentation and project handover processes would ensure smooth transitions. Proper documentation helps future team members understand ongoing work and enables interns to reflect on their experiences, creating valuable portfolios for their professional growth.

CHAPTER 9
CONCLUSION

The real estate price prediction project undertaken by TechCiti Software Pvt. Ltd. represents a significant step toward harnessing the transformative power of data science to address one of the most pressing challenges in the property sector—accurate and data-driven property valuation. By combining rich, diverse datasets with advanced analytical techniques, the project not only promises to deliver reliable price predictions but also offers actionable insights that can reshape how real estate stakeholders make informed decisions.

Throughout the project, we have meticulously navigated the data science lifecycle, from data collection and cleaning to feature engineering, model development, evaluation, and deployment. This holistic approach ensures that the final solution is both robust and adaptable to real-world market dynamics. The integration of cutting-edge machine learning algorithms, coupled with a deep understanding of the real estate domain, underscores our commitment to creating a tool that is not just technically sound but also practically impactful.

Moreover, the project reaffirms TechCiti's mission to drive innovation through data and technology, setting a new standard for how businesses can leverage predictive analytics to optimize decision-making. By providing stakeholders with accurate, real-time property valuations, the solution empowers them to navigate market fluctuations confidently and strategically.

In conclusion, this project marks a transformative journey toward data-driven real estate decision-making. It sets the stage for further advancements in predictive analytics and lays a solid foundation for future enhancements, such as incorporating live market data streams and integrating broader economic indicators. Through this initiative, TechCiti Software Pvt. Ltd. reinforces its vision of redefining industries through intelligent, data-centric solutions, setting the benchmark for excellence in real estate analytics.